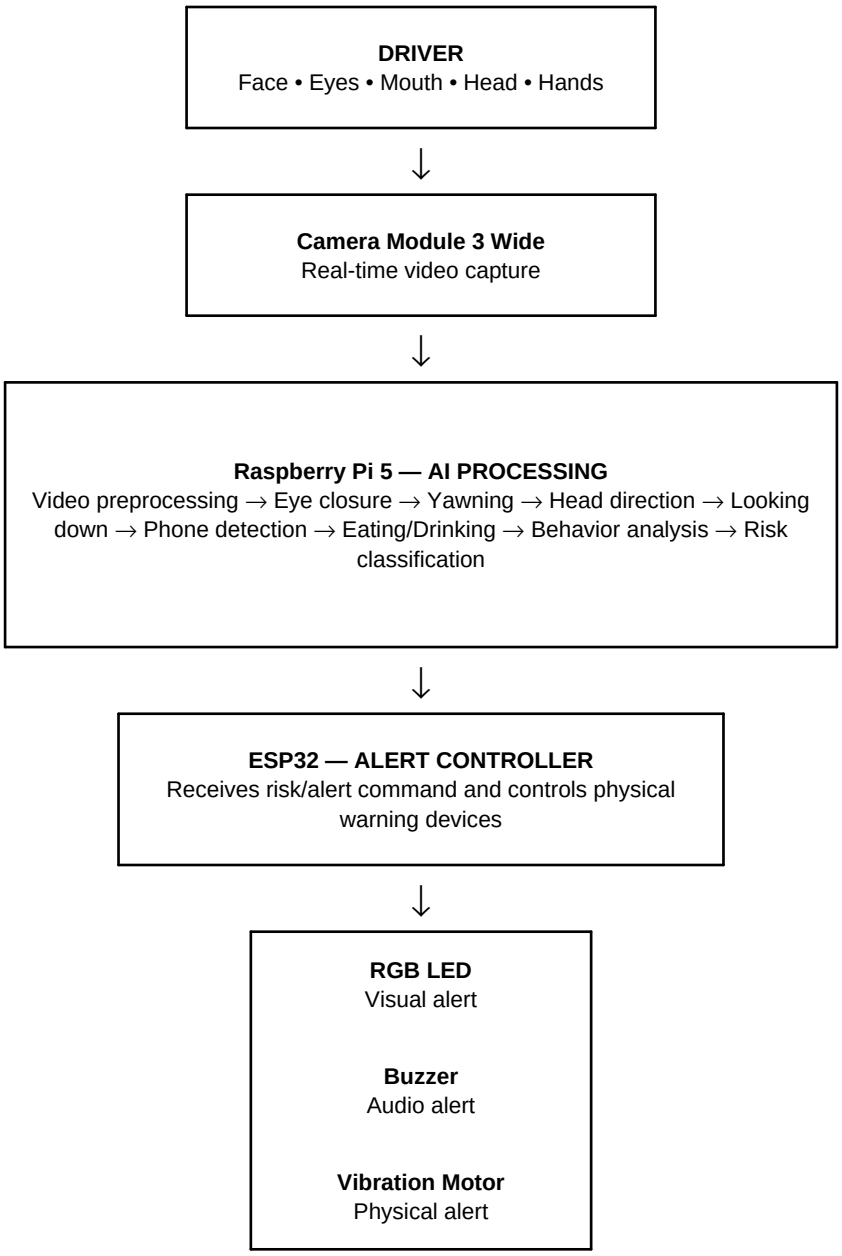
Portable AI-Based Driver Drowsiness and Distraction Detection System

GROUP NAME: AUTONOMOUS

Project Architecture — System Architecture + UML Activity Diagram

1. System Architecture Diagram

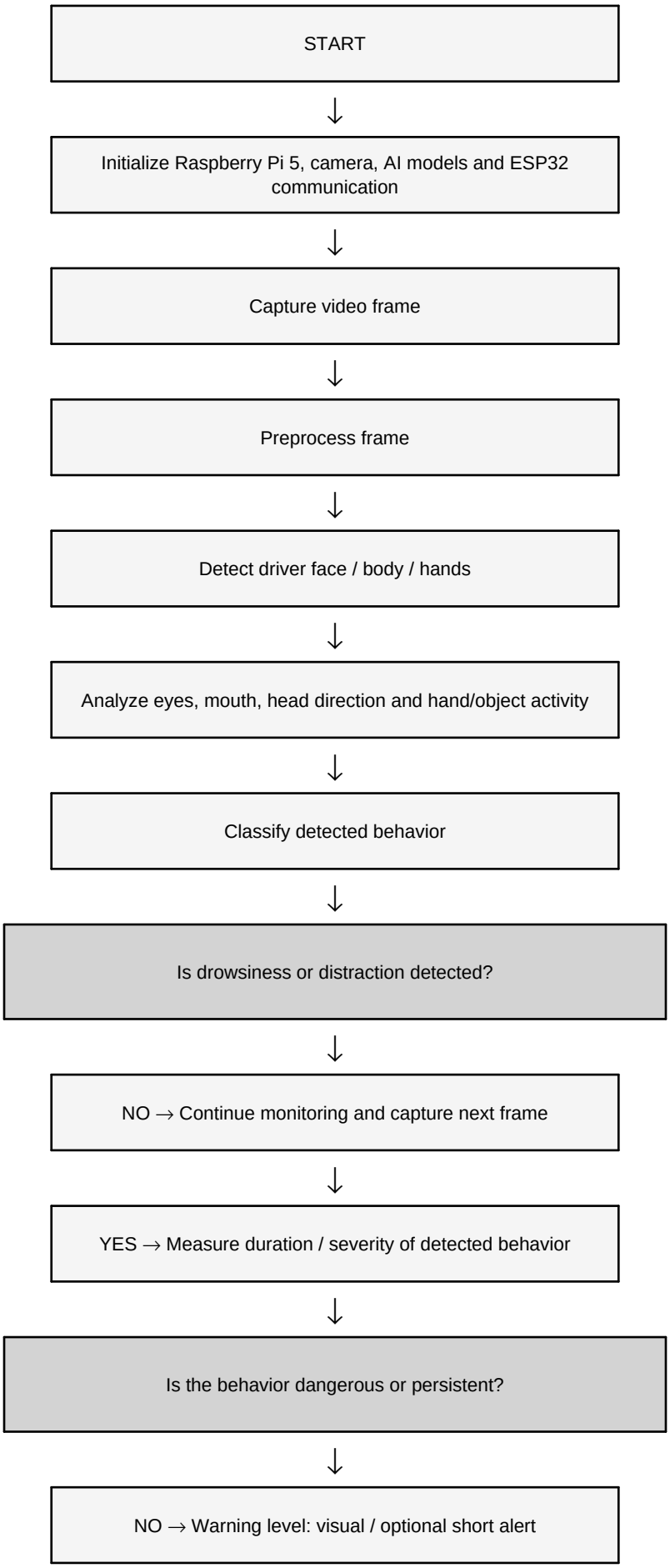


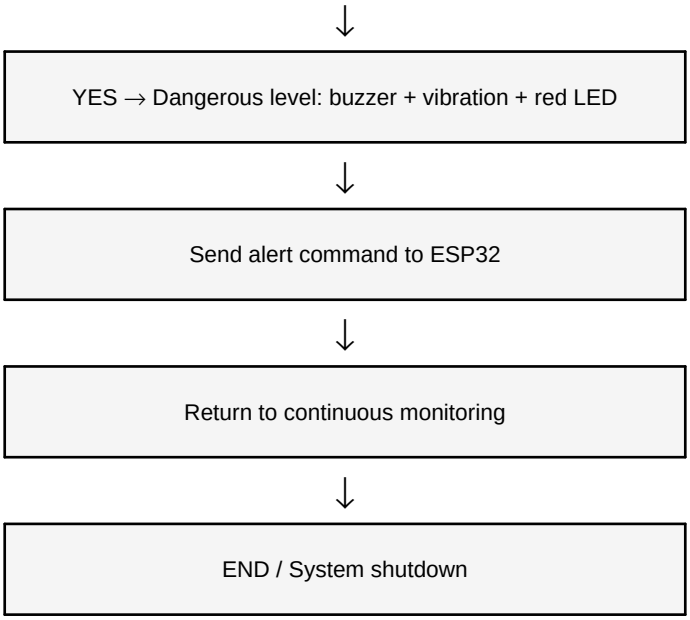
Optional Extension	**GPS Module** — location/event logging (only if implemented)
Power	USB-C power bank / regulated supply for Raspberry Pi 5, ESP32 and peripherals

System flow: Driver → Camera → Raspberry Pi 5 → AI detection and behavior analysis → Risk classification → ESP32 → RGB LED / Buzzer / Vibration Motor.

2. UML Activity Diagram

The activity flow below represents the continuous real-time operation of the system.





Implementation note: The activity diagram is intentionally independent of a specific AI framework. Your final implementation can use the computer-vision/AI models selected for face, eye, pose, hand, and object detection while keeping the same system-level workflow.

Architecture Notes for the Capstone Report

Layer	Purpose
Input Layer	Camera Module 3 Wide continuously captures the driver's upper-body view.
Processing Layer	Raspberry Pi 5 performs video processing, AI inference and behavior analysis.
Decision Layer	Detected behaviors are combined and classified into normal, warning or dangerous states.
Control Layer	ESP32 receives the alert command and controls the physical warning devices.
Output Layer	RGB LED, buzzer and vibration motor provide visual, audible and physical alerts.
Optional Extension	GPS and event logging can be added if the final prototype implements them.

Recommended report wording: “The proposed system is a portable embedded AI architecture in which the camera provides real-time driver video, the Raspberry Pi 5 performs multimodal behavior detection and risk classification, and an ESP32 drives physical alerts based on the detected risk level.”